Frequency Response-Design Specifications-Summary

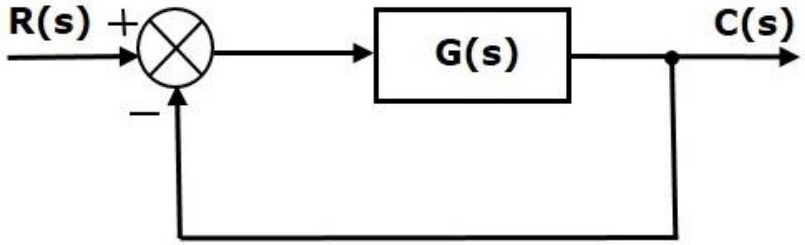
The frequency response parameters that we need to watch to design and conclude on the control system

Objectives

- Understand and interpret the Complementary Sensitivity frequency response as part of the specifications of a control system.
- Understand and interpret the Sensitivity frequency response as part of the specifications of a control system.
- Understand and interpret the phase margin as part of the specifications of a control system.
- Understand and interpret the gain margin as part of the specifications of a control system

1) Closed Loop Magnitude Frequency Response (Complementary Sensitivity)

- Consider the unity closed loop



- The closed loop transfer function is

$$T(s) = \frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)}$$

1) Closed Loop Magnitude Frequency Response (Complementary Sensitivity)

- The transfer function $T(s)$ is also called the complementary sensitivity.
- The frequency response of $T(s)$ is given by $T(j\omega)$. Consider now just the magnitude of $T(j\omega)$, $|T(j\omega)|$, and assume that there are no poles of $T(s)$ in the RHP (all poles have a non-zero negative real part).
- In these conditions, one way to appreciate the goodness of the control system is by looking at $|T(j\omega)|$ and compare with what we know about the canonical second order magnitude frequency response.

1) Closed Loop Magnitude Frequency Response (Complementary Sensitivity)

- First assume that $G(s)$ is type 1 or higher (that is $G(s)$ has at least one integrator), then

$$\lim_{s\to 0}|G(s)| = \infty \Rightarrow \lim_{\omega\to 0}|G(j\omega)| = \infty$$

- Then at low frequency $|G(s\to 0)| \gg 1$:

$$T(s\to 0) \approx \frac{G(s)}{G(s)} = 1 = 0_{\mathrm{db}}$$

- At high frequency, we are going to assume that $G(s)$ is proper, that is, there are more poles of $G(s)$ than zeros of $G(s)$, or in other words, the order of the denominator of $G(s)$ is higher than the order of the numerator of $G(s)$.

1) Closed Loop Magnitude Frequency Response (Complementary Sensitivity)

- In such a case, we have:

$$\lim_{s\to\infty} |G(s)| = 0$$

- Then, at high frequency

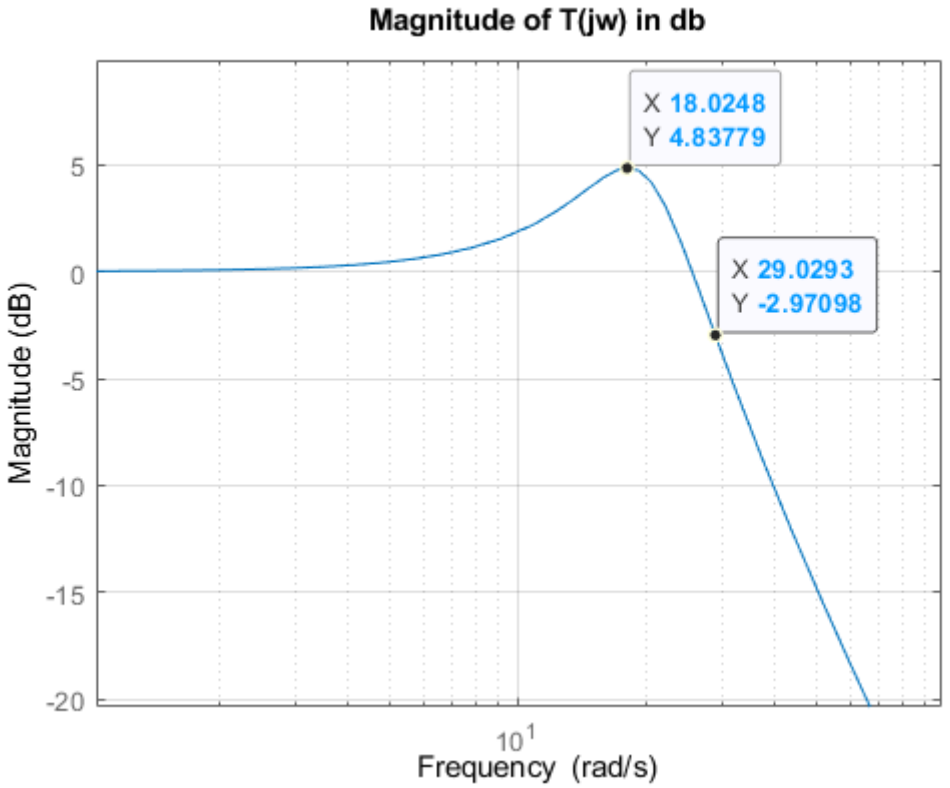
$$|G(s \to \infty)| \ll 1$$

$$T(s \to \infty) \approx \frac{G(s)}{1} = G(s)$$

- Which means that at high frequency $T(s)$ and $G(s)$ are very close.
- Assuming again that all the assumptions are given, we can look at the magnitude of the Bode of $T(j\omega)$ and conclude if the control system is good or not.

1) Closed Loop Magnitude Frequency Response (Complementary Sensitivity)

- A typical magnitude Bode of $T(jw)$ would look like this:



1) Closed Loop Magnitude Frequency Response (Complementary Sensitivity)

- To determine the goodness of the control system, first, find the resonant coefficient M_r:

$$M_{r(\mathrm{db})} = \mathrm{Peak\ value}_{\mathrm{db}} - \mathrm{Low\ Freq\ Gain}_{\mathrm{db}}$$

- From our Bode of the previous slide,

$$M_{r(\mathrm{db})} = 4.84\ \mathrm{db} - 0\ \mathrm{db} = 4.84\ \mathrm{db}$$

- Based on M_r, you could estimate the damping coefficient ζ related to this M_r. The damping coefficient will allow you to conclude about the overshoot, oscillations and relative stability. Typically a good M_r is less than 3 db.
- Second, in the next slide.

1) Closed Loop Magnitude Frequency Response (Complementary Sensitivity)

- Second, you can determine the bandwidth of your control system. The bandwidth ω_{BW} is the frequency when the magnitude of $T(jw)$ is 3 db below the low frequency gain.
- The low frequency gain is 0 db which tells us that our assumption that $G(s)$ was type 1 or higher is correct.
- Since the low frequency gain is 0 db, we need to find directly the frequency when the magnitude is -3 db. In the previous Bode, we find:

$$\omega_{BW} \approx 29.03\, rad/s$$

- A high value of ω_{BW} will tell you that your control system responds fast but could be susceptible to noise as more noise will pass through. Conversely, a low value of ω_{BW} will tell you that your control system responds slowly but is less sensitive to noise. Your application will determine what bandwidth is good.

1) Closed Loop Magnitude Frequency Response (Complementary Sensitivity)

- Let's consider now the case of $G(s)$ being type 0.
- If $G(s)$ is type 0, we need to take into account now that:

$$G(0) = K > 0$$

- Therefore,

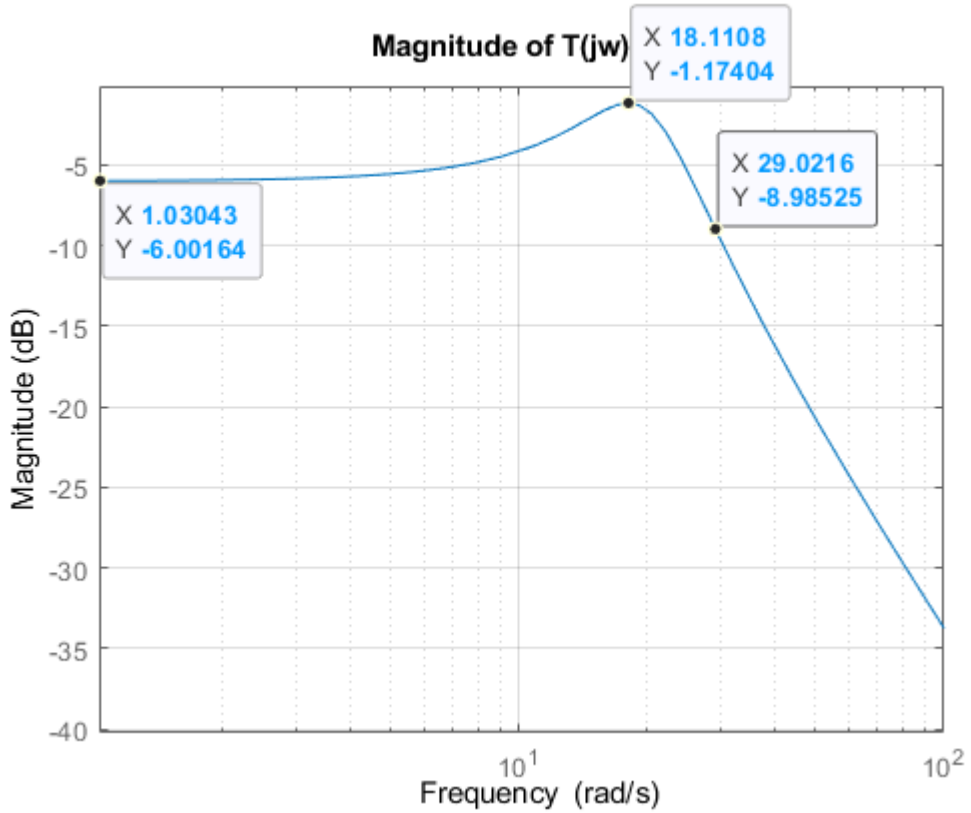
$$T(0) = \frac{G(0)}{1 + G(0)} = \frac{K}{1 + K} < 1$$

- And,

$$|T(j0)|_{\text{db}} = 20 \log\left(\frac{K}{1 + K}\right) < 0 \text{ db}$$

1) Closed Loop Magnitude Frequency Response (Complementary Sensitivity)

- A typical Bode magnitude of $T(j\omega)$ when $G(s)$ is type zero looks like this:



1) Closed Loop Magnitude Frequency Response (Complementary Sensitivity)

- In this case, note that

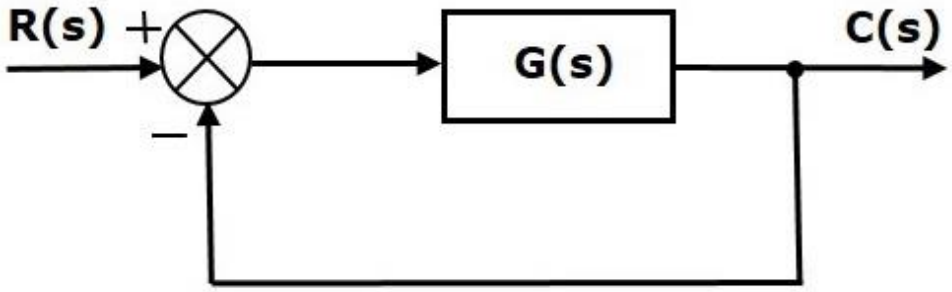
$$M_r \approx -1.175\ \text{db} - (-6.002\ \text{db}) = 4.827\ \text{db}$$

- And, to find the bandwidth we need to find the magnitude which is 3 db below -6 db (low frequency gain), that is -9 db:

$$\omega_{BW} \approx 29.022\ \text{rad/s}$$

2) Sensitivity Function

- In a unity feedback controls system, the sensitivity function is given by
  $G(s) = \frac{1}{1+G(s)}$
- Where $G(s)$ is the direct forward open loop transfer function:



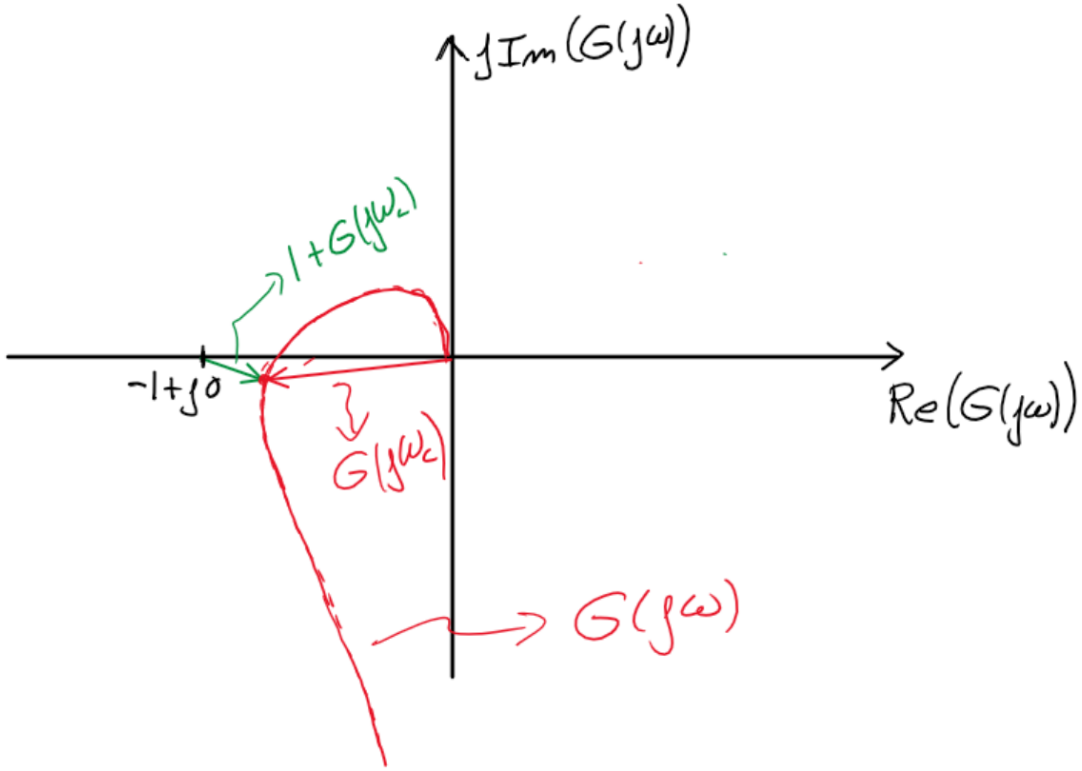
2) Sensitivity Function

- The Nyquist of $G(s)$ shows the importance of the quantity $1 + G(s)$ (or equivalently $1 + G(j\omega)$ if we consider the frequency response)
- Assume as we did before, that all the poles of T(s), that is, all the zeros of 1+G(s), are in the RHP as before (closed loop system is stable), then, we can conclude about the goodness of the control system as follows.
- The magnitude of $(1 + G(j\omega))$, that is, $|1 + G(j\omega)|$is an indication of how close $G(j\omega)$ Nyquist is to $-1 + j0$. The closer the Nyquist of $G(j\omega)$ is to $-1 + j0$, the worse the closed loop control system is.

2) Sensitivity Function

- Note the Nyquist of $G(j\omega)$. At certain frequency ω_c, $|1 + G(j\omega_c)|$ reaches its minimum value. This is the closest point of $G(j\omega)$ to $-1 + j0$. We want this closest point to be far away enough of $-1 + j0$; if this is the case, the control system will be good and robust.
- In other words:
  $\min_{\omega} |1 + G(j\omega)| \geq l$ (lower limit)

How to choose l? It depends on the application.



2) Sensitivity Function

- However we don't work with $1 + G(j\omega)$ but with the sensitivity $S(j\omega)$, that is,

$$S(j\omega) = \frac{1}{1 + G(j\omega)}$$

- Which is the reciprocal of $1 + G(j\omega)$, therefore, to guarantee robustness of a control system, we need to put conditions on $S(j\omega)$:

$$\max_{\omega} |S(j\omega)| = \max_{\omega} \left| \frac{1}{1 + G(j\omega)} \right| \leq L \text{ (Upper Limit)}$$

The smaller L, the better. What is the value of L? It depends on the application. Typically, $L = 6$ db is used as reference

2) Sensitivity Function

- Conclusion: the highest value of $|S(j\omega)|$ should be low enough to guarantee robustness of a stable control system.
- Notice that:

$$T(s) + S(s) = 1$$

- We can demonstrate that the sensor disturbance (sensor noise) is directly affected by the sensitivity. The noise is reduced/amplified by $|S(j\omega)|$.

$|S(j\omega)| < 1 \Rightarrow$ disturbances with frequency ω are reduced by controller

$|S(j\omega)| > 1 \Rightarrow$ disturbances with frequency ω are magnified by controller

2) Sensitivity Function

- Let's consider the following example. We have a plant given by

$$G_p(s) = \frac{4}{s(s+2)}$$

- Engineer A designed the following controller:

$$G_{cA}(s) = \frac{41.7(s+4.41)}{(s+18.4)}$$

- Engineer B designed the following controller:

$$G_{cB}(s) = \frac{41.7(s+7)}{(s+18.4)}$$

- a) What is the more robust design? b) What system is the fastest? Both controllers produce stable systems. See solution in *SensitivityProblem.pdf*

3) Phase Margin

- **Phase Margin**: The phase margin is that amount of additional phase lag at the **gain crossover frequency** (O db frequency or w_{0db}) required to bring the (closed loop) system to the verge of instability. The **gain crossover frequency** is the frequency at which $|G(jw)| = 1$ (or 0 db). The phase margin is then:

$$PM = 180 + \angle G(jw_{0db})$$

- We say that the (closed loop) system is stable if $PM > 0$, and unstable if $PM \leq 0$
- Good PM between 40 and 60 degrees.
- A PM by itself is not enough. PM needs to be verified with the gain margin (more accurately, we need to check it also with $|S(j\omega)|$)

4) Gain Margin

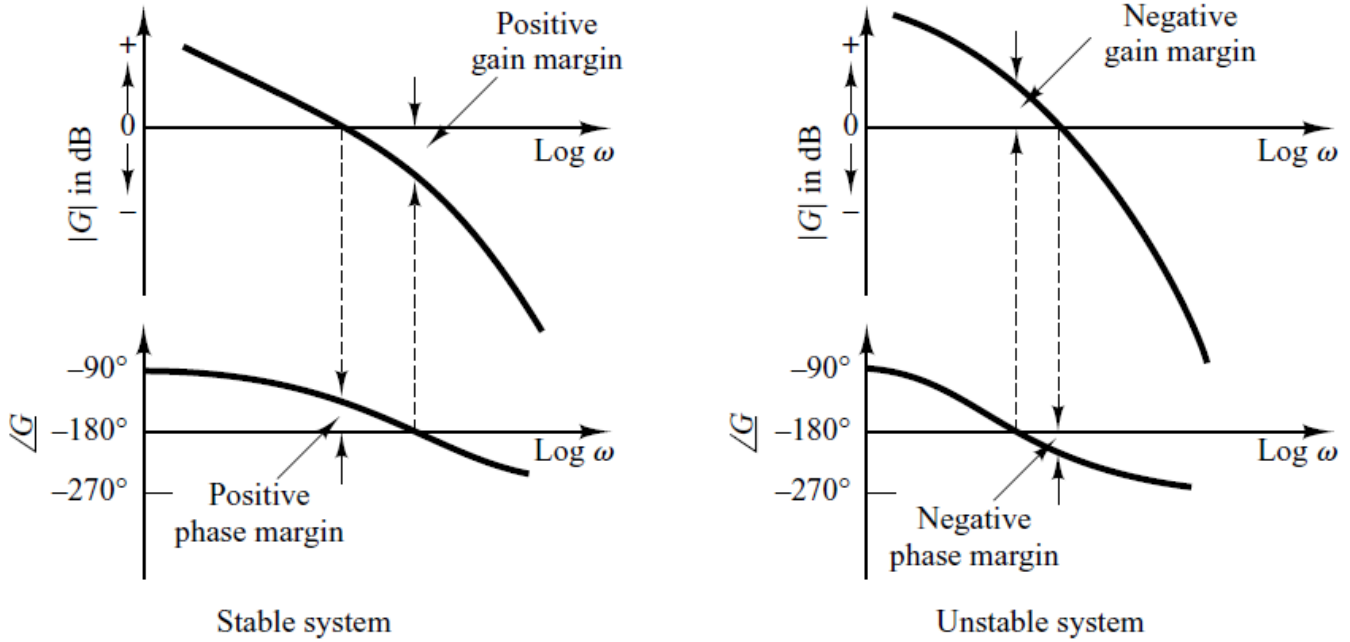
- **Gain Margin**: The gain margin is the reciprocal of the magnitude |G(jw)| at the frequency at which the phase angle is -180, this frequency is known as the **phase crossover frequency** (w_{-180}), then the gain margin is given by

$$GM = \frac{1}{|G(jw_{-180})|}$$

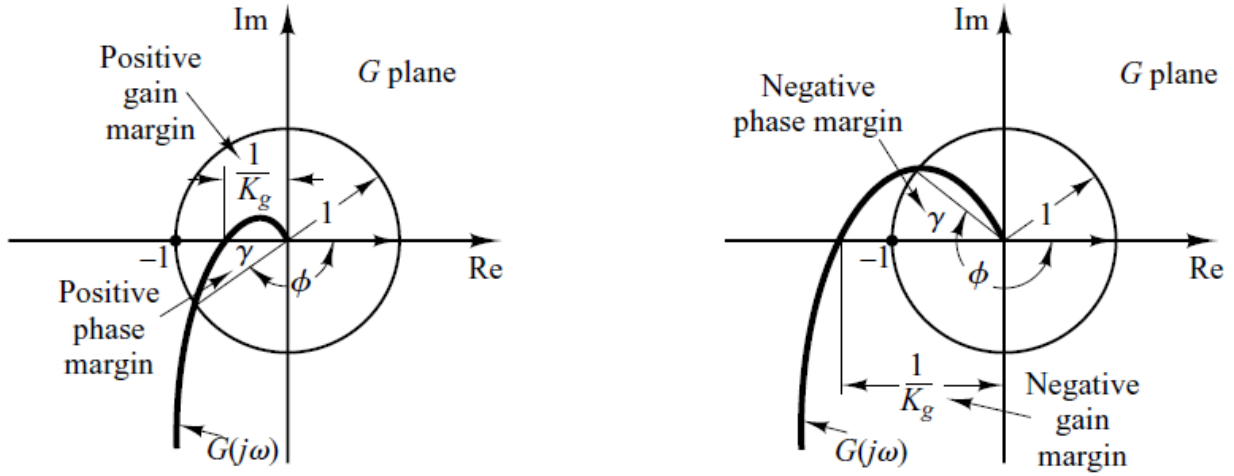
$$GM_{db} = -20\log|G(jw_{-180})| = -|G(jw_{-180})|_{db}$$

- We say that the closed loop system is stable if GM>1 ($GM_{db} > 0$ db) and unstable if $GM \leq 1$ ($GM_{db} \leq 0$ db)
- Good GM bigger or equal than 6db
- A GM by itself is not enough. GM needs to be verified with the phase margin (more accurately, we need to check it also with $|S(j\omega)|$)

Stability Margins



(a)



(b)